Muscle Coactivation Levels Reflect a Tradeoff Between Metabolic Cost and Performance When Responding to Physical Disturbances During Upper Limb Posture Control

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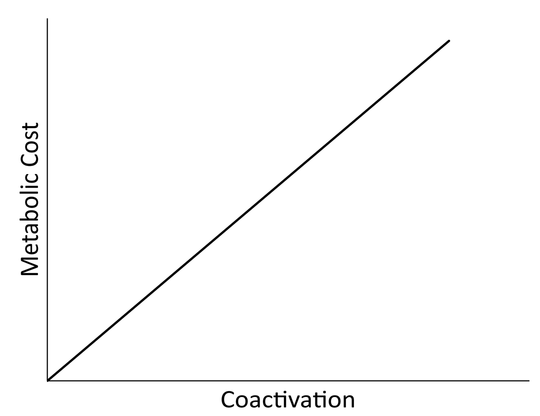
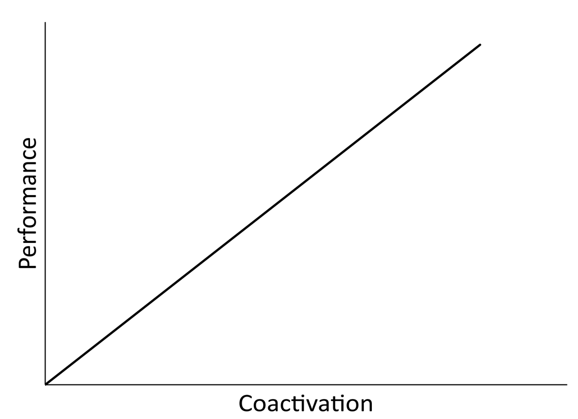
Introduction

Muscle coactivation - the simultaneous activity of agonist and antagonist muscles - has been shown to improve motor performance by enabling muscles to generate forces more rapidly and respond more effectively to sensory feedback. However, this enhanced performance comes at a cost, and the metabolic expense associated with muscle coactivation remains unknown. The main objective of this study is to investigate the trade-off between metabolic cost and performance under different levels of muscle coactivation.

Hypothesis and Prediction

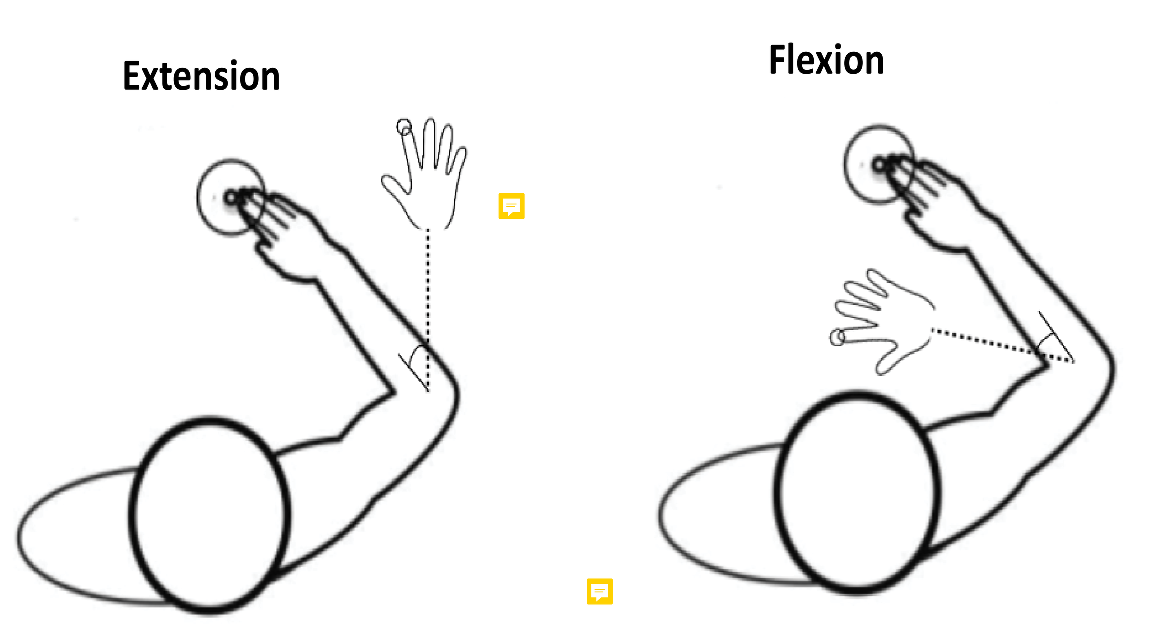
We hypothesize that there is a trade-off between muscle coactivation, metabolic cost, and task performance. We predict that higher levels of coactivation will lead to improved performance at the expense of increased metabolic cost.





Methods

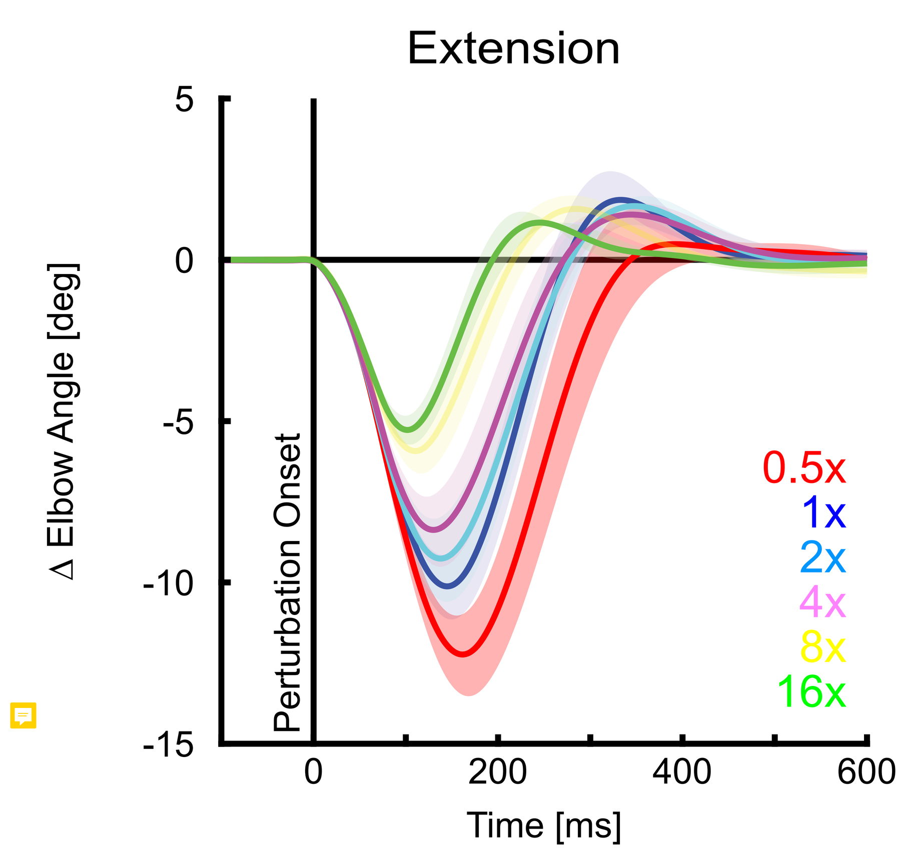
Five healthy participants attempted to maintain a fixed arm posture in a Kinarm exoskeleton robot while encountering mechanical disturbances.



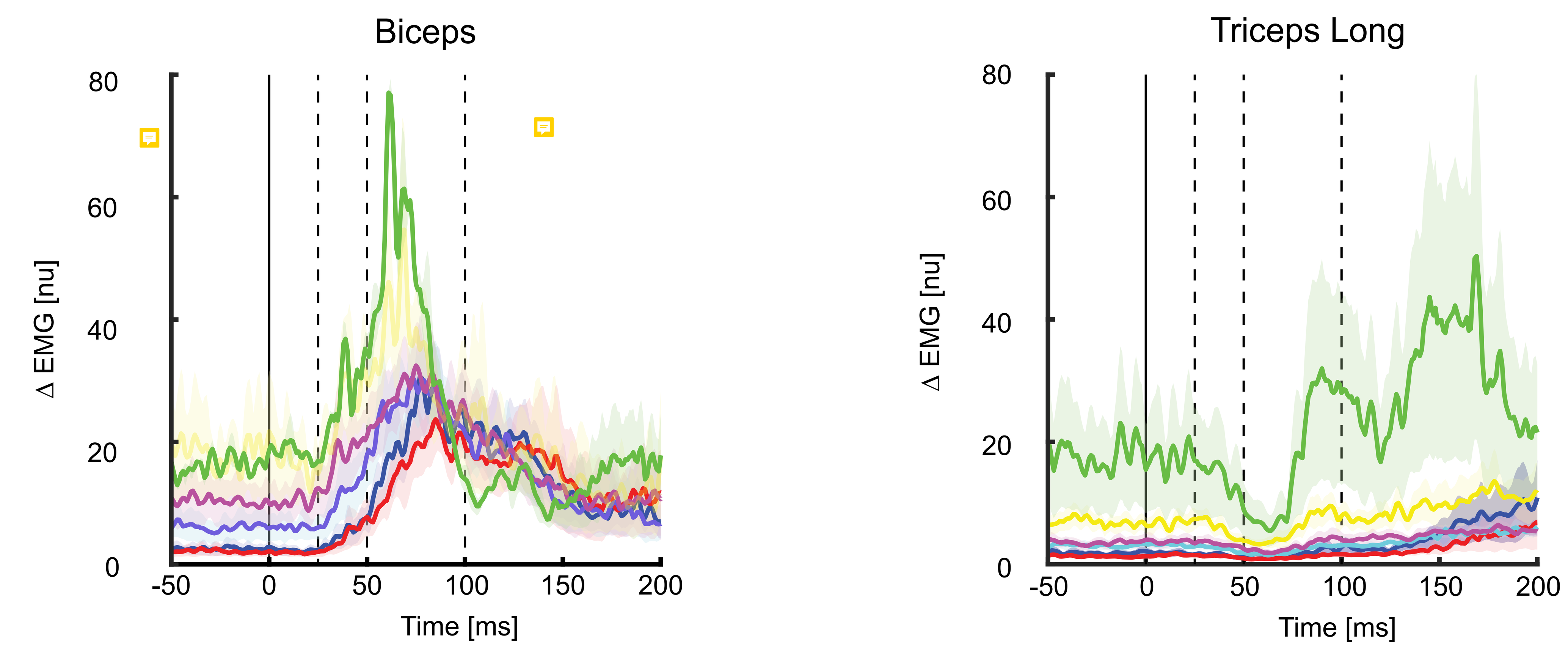
The disturbances rapidly extended or flexed the elbow and displaced the participant's arm from the target. They were required to return to the target within 400 ms. Participants first performed the task under self-selected levels of muscle coactivation. Biofeedback was then used to maintain a level of coactivation corresponding to [0.5, 2, 4, 8, 16]x their self-selected (1x) level. Metabolic cost was measured with gas respirometry, the activity of upper limb muscles using surface EMG, and behavioral responses using success rates.

Results

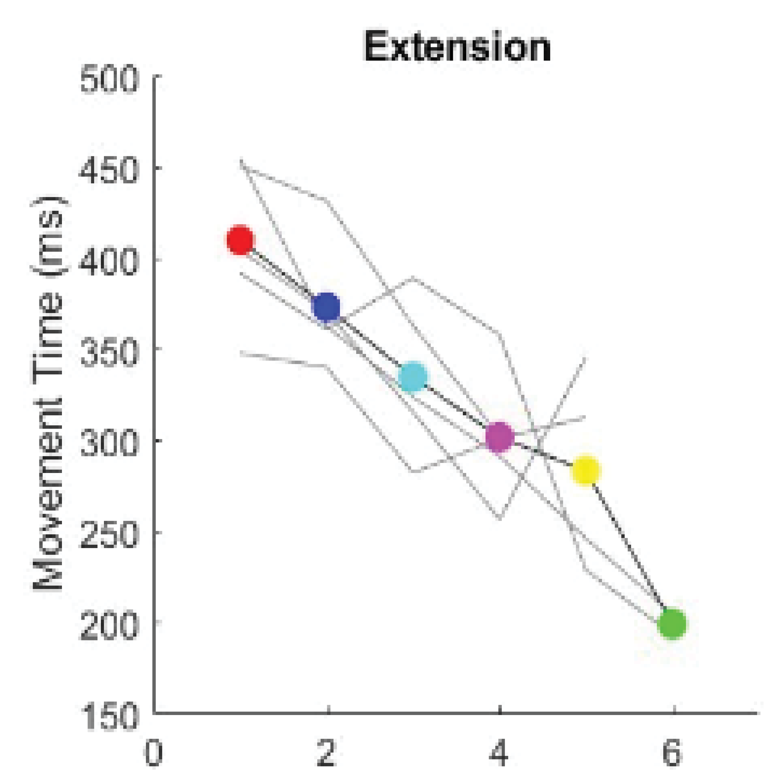
Muscle Coactivation Reduces Peak Displacement



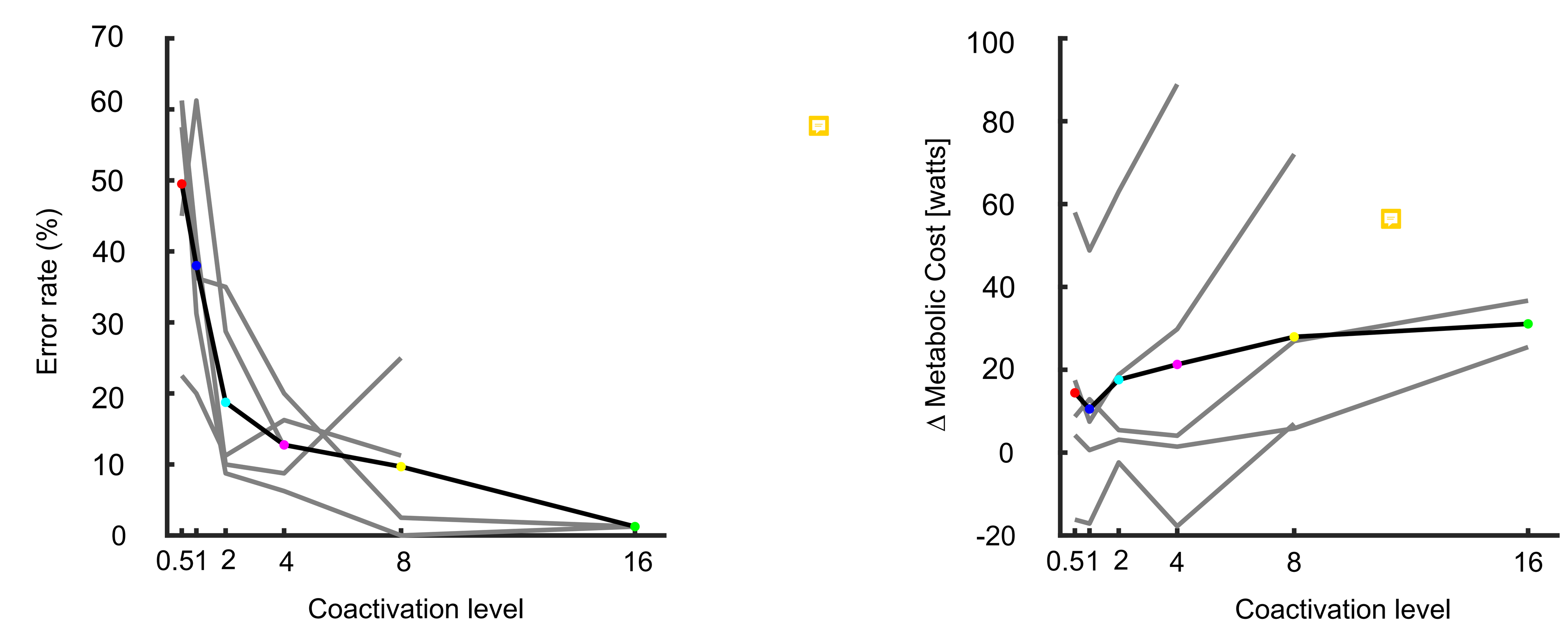
Coactivation Increases Agonist Activity and Reduces Antagonist Response



Higher Coactivation Levels Allows For Quicker Response



High Coactivation Improves Performance But Increases Metabolic Cost



Discussion

Spontaneous levels of muscle coactivation seem to prioritize the minimization of metabolic cost at the expense of performance. However, increasing the level of coactivation via biofeedback-controlled muscle activity improves performance at the expense of metabolic cost. This research offers insights into the role of muscle coactivation in responding to sensory feedback.

Conclusion

Participants naturally prioritize metabolic cost over performance when selecting muscle coactivation levels, highlighting the complex interplay between metabolic cost and performance in human movement optimization.